Web Development Activity Based Learning

Project Title: Online News Portal

Description

The Online News Portal (GlobalNews) is a responsive news website built using HTML and CSS. The home page presents a featured headline over an image overlay, a breaking-news ticker, and category-wise news sections (National, International, Sports, Technology and Business) displayed as article cards, along with a sidebar of featured stories, trending topics and author information. The project uses semantic HTML tags, an external stylesheet, CSS Grid and Flexbox layouts, a sticky navigation bar, colour-coded category labels, hover effects and media queries so that the page adapts to computers, tablets and mobile phones.

1. Introduction

This activity introduces the two core technologies of the web: **HTML** (HyperText Markup Language), which defines the content and structure of a page, and **CSS** (Cascading Style Sheets), which controls how that content looks. The project is the home page of an online news portal called **GlobalNews**. The HTML (*news.html*) and the CSS (*news.css*) are kept in two separate files and connected with a **link** tag. A news website is a good learning project because it combines many layout ideas: a menu, a large headline, repeated article cards, a side column and a footer.

Objectives

- To structure a news page using semantic HTML tags such as header, nav, main, section, article, aside and footer.
- To link an external stylesheet so that content and styling are kept separate.
- To build a two-column page layout (main news area and sidebar) using CSS Grid.
- To create repeated article cards and colour-coded category labels with reusable CSS classes.
- To add a sticky navigation bar, an image overlay, hover effects and a responsive layout using media queries.

Website Sections at a Glance

Section	Purpose
Top Bar	Shows the date and the social media names.
Navigation Bar	Sticky bar with the GlobalNews logo and links to the news categories.
Featured Headline	Large top story over a darkened background image, with a "Read Full Story" button.
Breaking News	A red label and a scrolling text ticker.
Category Navigation	A row of links that jumps to each news section.
News Sections	National, International, Sports, Technology and Business, each with two article cards.
Sidebar	Featured Stories, a numbered Trending Now list and an Our Author panel.
Footer	Site name, tagline, links and the copyright line.

2. HTML Source Code (news.html)

The screenshots below show the HTML file in the code editor, followed by the complete source listing.

Source Code in the Editor

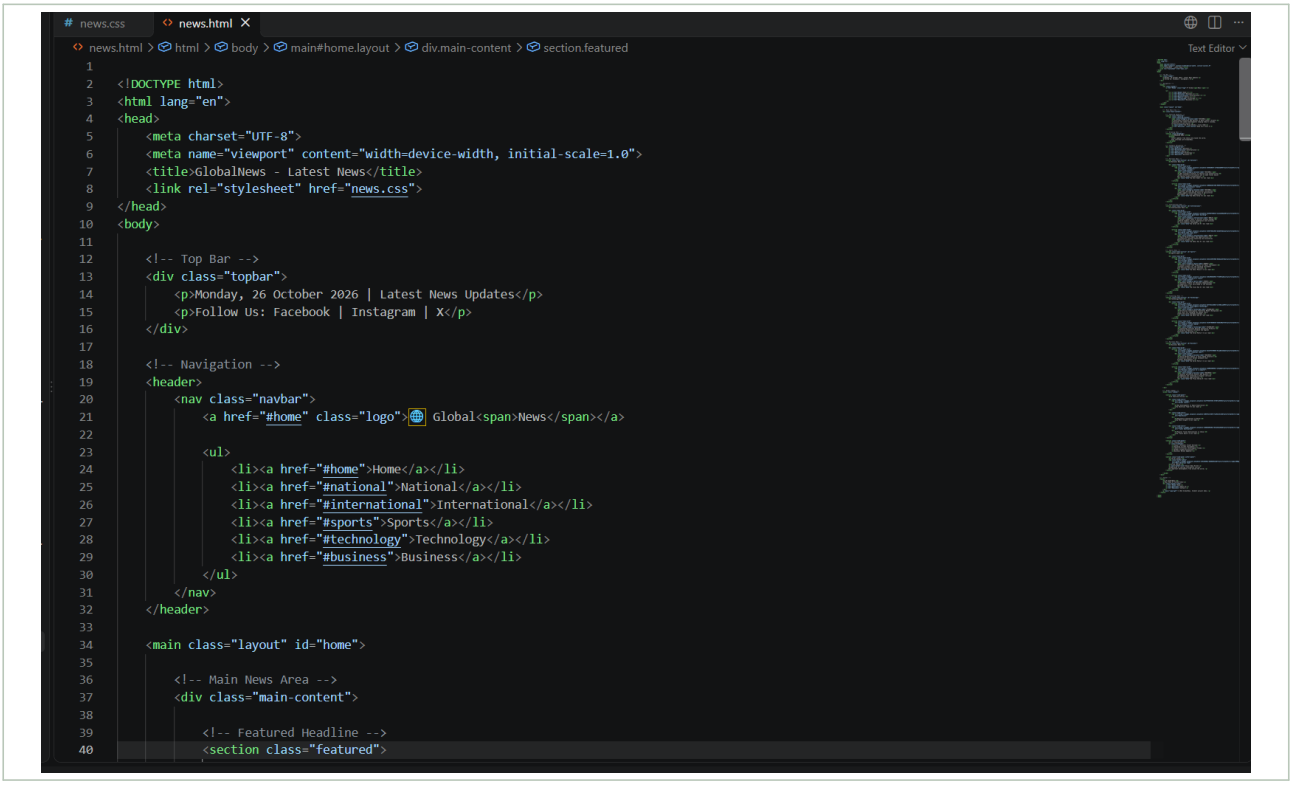


Figure 1: news.html in the code editor - document head, top bar, navigation and start of the main layout (lines 1-40)

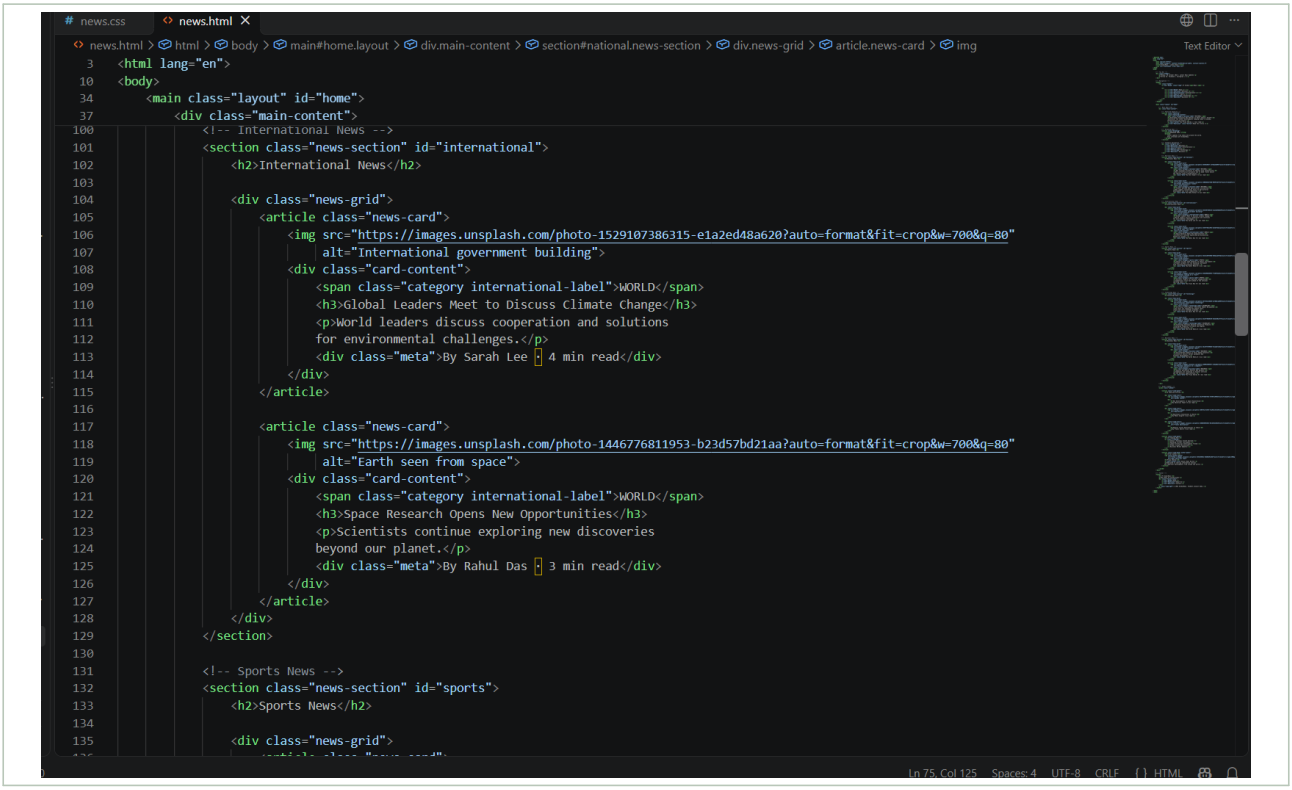


Figure 2: news.html in the code editor - International News section with article cards (lines 100-135)

Complete Source Listing

Note: emoji and some symbols in the original file are shown here as HTML character codes (for example 🌐 stands for the globe emoji), because the PDF code font cannot draw them. The browser displays them normally.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>GlobalNews - Latest News</title>
    <link rel="stylesheet" href="news.css">
</head>
<body>

    <!-- Top Bar -->
    <div class="topbar">
        <p>Monday, 26 October 2026 | Latest News Updates</p>
        <p>Follow Us: Facebook | Instagram | X</p>
    </div>

    <!-- Navigation -->
    <header>
        <nav class="navbar">
            <a href="#home" class="logo">&#127760; Global<span>News</span></a>

            <ul>
                <li><a href="#home">Home</a></li>
                <li><a href="#national">National</a></li>
                <li><a href="#international">International</a></li>
                <li><a href="#sports">Sports</a></li>
                <li><a href="#technology">Technology</a></li>
                <li><a href="#business">Business</a></li>
            </ul>
        </nav>
    </header>

    <main class="layout" id="home">

        <!-- Main News Area -->
        <div class="main-content">
```

```
<!-- Featured Headline -->
<section class="featured">
    <div class="featured-content">
        <span class="category national-label">NATIONAL</span>
        <h1>Government Announces New Reforms to Boost India's Growth</h1>
        <p>Discover the latest developments shaping India's economy,
        infrastructure and future.</p>
        <p class="author">By Priya Sharma | 4 min read</p>
        <a href="#national" class="button">Read Full Story →</a>
    </div>
</section>

<!-- Breaking News -->
<section class="breaking">
    <strong>BREAKING NEWS</strong>
    <marquee>
        Latest updates from India and around the world.
        Stay informed with GlobalNews.
    </marquee>
</section>

<!-- Category Navigation -->
<div class="category-nav">
    <a href="#national">National</a>
    <a href="#international">International</a>
    <a href="#sports">Sports</a>
    <a href="#technology">Technology</a>
    <a href="#business">Business</a>
</div>

<!-- National News -->
<section class="news-section" id="national">
    <h2>National News</h2>

    <div class="news-grid">
        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1544620347-c4fd4a3d5957?auto=f
                ormat&fit=crop&w=700&q=80"
                 alt="Public transport">
            <div class="card-content">
                <span class="category national-label">NATIONAL</span>
                <h3>New Transport Projects to Improve City Connectivity</h3>
                <p>New infrastructure projects aim to make travel easier
                and improve urban transportation.</p>
                <div class="meta">By Anil Kumar · 3 min read</div>
            </div>
        </article>

        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1500382017468-9049fed747ef?aut
                o=format&fit=crop&w=700&q=80"
                 alt="Green agricultural fields">
            <div class="card-content">
                <span class="category national-label">NATIONAL</span>
                <h3>Farmers Welcome New Agricultural Initiatives</h3>
                <p>New support initiatives focus on agricultural
                development and rural communities.</p>
                <div class="meta">By Neha Verma · 3 min read</div>
            </div>
        </article>
    </div>
</section>

<!-- International News -->
<section class="news-section" id="international">
    <h2>International News</h2>

    <div class="news-grid">
        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1529107386315-e1a2ed48a620?aut
                o=format&fit=crop&w=700&q=80"
                 alt="International government building">
            <div class="card-content">
                <span class="category international-label">WORLD</span>
                <h3>Global Leaders Meet to Discuss Climate Change</h3>
                <p>World leaders discuss cooperation and solutions
```

```
                for environmental challenges.</p>
                <div class="meta">By Sarah Lee · 4 min read</div>
            </div>
        </article>

        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1446776811953-b23d57bd21aa?aut
                o=format&fit=crop&w=700&q=80"
                 alt="Earth seen from space">
            <div class="card-content">
                <span class="category international-label">WORLD</span>
                <h3>Space Research Opens New Opportunities</h3>
                <p>Scientists continue exploring new discoveries
                beyond our planet.</p>
                <div class="meta">By Rahul Das · 3 min read</div>
            </div>
        </article>
    </div>
</section>

<!-- Sports News -->
<section class="news-section" id="sports">
    <h2>Sports News</h2>

    <div class="news-grid">
        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1531415074968-036ba1b575da?aut
                o=format&fit=crop&w=700&q=80"
                 alt="Cricket match">
            <div class="card-content">
                <span class="category sports-label">SPORTS</span>
                <h3>Indian Cricket Team Prepares for Major Tournament</h3>
                <p>Players prepare for the upcoming tournament
                with training and practice sessions.</p>
                <div class="meta">By Rohit Menon · 2 min read</div>
            </div>
        </article>

        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1461896836934-ffe607ba8211?aut
                o=format&fit=crop&w=700&q=80"
                 alt="Athletes running on a track">
            <div class="card-content">
                <span class="category sports-label">SPORTS</span>
                <h3>Athletes Prepare for National Championships</h3>
                <p>Competitors train hard ahead of the national
                sporting event.</p>
                <div class="meta">By Arjun Rao · 3 min read</div>
            </div>
        </article>
    </div>
</section>

<!-- Technology News -->
<section class="news-section" id="technology">
    <h2>Technology News</h2>

    <div class="news-grid">
        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1677442136019-21780ecad995?aut
                o=format&fit=crop&w=700&q=80"
                 alt="Artificial intelligence technology">
            <div class="card-content">
                <span class="category technology-label">TECHNOLOGY</span>
                <h3>Artificial Intelligence Transforms Modern Workplaces</h3>
                <p>AI tools are changing how people learn,
                create and solve everyday problems.</p>
                <div class="meta">By Aditi Rao · 4 min read</div>
            </div>
        </article>

        <article class="news-card">
            <img src="https://images.unsplash.com/photo-1518770660439-4636190af475?aut
                o=format&fit=crop&w=700&q=80"
                 alt="Computer circuit board">
            <div class="card-content">
```

```
                    <span class="category technology-label">TECHNOLOGY</span>
                    <h3>Technology Startups Develop Innovative Products</h3>
                    <p>Emerging companies introduce new digital
                    solutions and services.</p>
                    <div class="meta">By Kiran Mehta · 3 min read</div>
                </div>
            </article>
        </div>
    </section>

    <!-- Business News -->
    <section class="news-section" id="business">
        <h2>Business News</h2>

        <div class="news-grid">
            <article class="news-card">
                <img src="https://images.unsplash.com/photo-1611974789855-9c2a0a7236a3?aut
                    o=format&fit=crop&w=700&q=80"
                     alt="Stock market financial chart">
                <div class="card-content">
                    <span class="category business-label">BUSINESS</span>
                    <h3>Stock Markets Remain in Focus Among Investors</h3>
                    <p>Investors monitor market movements and
                    economic developments.</p>
                    <div class="meta">By Karan Mehta · 3 min read</div>
                </div>
            </article>

            <article class="news-card">
                <img src="https://images.unsplash.com/photo-1460925895917-afdab827c52f?aut
                    o=format&fit=crop&w=700&q=80"
                     alt="Business analytics on a computer">
                <div class="card-content">
                    <span class="category business-label">BUSINESS</span>
                    <h3>Digital Businesses Explore New Markets</h3>
                    <p>Companies are investing in digital services
                    and new business opportunities.</p>
                    <div class="meta">By Priya Sharma · 4 min read</div>
                </div>
            </article>
        </div>
    </section>

</div>

<!-- Right Sidebar -->
<aside class="sidebar">

    <section class="side-panel">
        <h2>&#11088; Featured Stories</h2>

        <div class="side-story">
            <img src="https://images.unsplash.com/photo-1517976487492-5750f3195933?auto=fo
                rmat&fit=crop&w=250&q=80"
                 alt="Rocket launch">
            <div>
                <h4>New Developments in Space Exploration</h4>
                <p>By Editorial Team · 3 min read</p>
            </div>
        </div>

        <div class="side-story">
            <img src="https://images.unsplash.com/photo-1505751172876-fa1923c5c528?auto=fo
                rmat&fit=crop&w=250&q=80"
                 alt="Healthcare">
            <div>
                <h4>Healthcare Innovations to Watch</h4>
                <p>By Meera Singh · 2 min read</p>
            </div>
        </div>

        <div class="side-story">
            <img src="https://images.unsplash.com/photo-1488646953014-85cb44e25828?auto=fo
                rmat&fit=crop&w=250&q=80"
                 alt="Travel destination">
            <div>
```

```
                    <h4>Popular Travel Destinations in India</h4>
                    <p>By Travel Desk · 4 min read</p>
                </div>
            </div>
        </section>

        <section class="side-panel">
            <h2>&#128293; Trending Now</h2>
            <ol class="trending">
                <li>India's Economic Growth Outlook</li>
                <li>Upcoming Cricket Tournament</li>
                <li>Latest Artificial Intelligence Trends</li>
                <li>Global Climate Discussions</li>
                <li>Business Market Updates</li>
            </ol>
        </section>

        <section class="side-panel author-panel">
            <h2>&#128100; Our Author</h2>
            <img class="author-photo"
                 src="https://images.unsplash.com/photo-1551836022-d5d88e9218df?auto=format&fi
                     t=crop&w=300&q=80"
                 alt="News editorial team">
            <h3>Priya Sharma</h3>
            <p class="author-role">Senior News Writer</p>
            <p>Dedicated to sharing informative stories
            and important developments from around the world.</p>
        </section>

    </aside>
</main>

<!-- Footer -->
<footer>
    <h2>&#127760; GlobalNews</h2>
    <p>Real News · Trusted Views</p>
    <div class="footer-links">
        <a href="#home">Home</a>
        <a href="#national">About Us</a>
        <a href="#business">Contact</a>
    </div>
    <p class="copyright">© 2026 GlobalNews. Student project demo.</p>
</footer>

</body>
</html>
```

3. Description of the HTML Code

Document Structure

The file begins with **<!DOCTYPE html>** and **<html lang="en">**. The **<head>** contains **<meta charset="UTF-8">** (so emoji and symbols display correctly), the **viewport** meta tag (so the page scales on mobile screens), the **<title>** "GlobalNews - Latest News" and **<link rel="stylesheet" href="news.css">**, which connects the external stylesheet. All visible content is inside **<body>**, and comments such as *<!-- Top Bar -->* label each part.

Tags and Attributes Used

Tag / attribute	Where it is used	Purpose
<header>, <nav>, <main>, <aside>, <footer>	Menu bar, page content, sidebar, page bottom	Semantic tags that describe the role of each area.
<section> and <article>	News sections and individual news cards	A section groups related content; an article is a self-contained news item.
, ,	Menu links; Trending Now list	Unordered list for the menu, ordered list for the numbered trending topics.
<h1> to <h4>	Top headline, section titles, card titles	Headings in decreasing order of importance.
<p>, , ,	Text, category labels, BREAKING NEWS label, containers	Paragraphs, inline text and general containers for grouping.
<marquee>	Breaking news ticker	Scrolls its text across the bar. It is an older HTML element, but browsers still support it.
<img> with src and alt	Article cards, sidebar stories, author photo	Displays a photo; alt gives a text description for accessibility.
<a href>	Menu, buttons, footer	Links. "#national" jumps to the section with id="national".
class attribute	topbar, navbar, news-card, category ...	Names an element so CSS can style it. An element can have several classes, such as "category national-label".
id attribute	home, national, international, sports, technology, business	Unique name that lets links jump to a section.

Section-by-Section Explanation

- **Top bar:** a div with two paragraphs: the date with "Latest News Updates" and the social media names.
- **Navigation:** a header holding a nav with the logo link (globe emoji and "GlobalNews") and a list of six links: Home, National, International, Sports, Technology and Business.
- **Main layout:** a <main> element with two children: "main-content" for the news and an <aside> for the sidebar. This two-part structure is what CSS Grid later turns into two columns.
- **Featured headline:** a label, the <h1> headline "Government Announces New Reforms to Boost India's Growth", a short paragraph, the author line and a "Read Full Story" button link.

- **Breaking news and category navigation:** a "BREAKING NEWS" label with a marquee, followed by five quick links to the sections.
- **News sections:** five sections, each with an <h2> and a "news-grid" of two <article> cards. Each card has an image, a coloured category label (NATIONAL, WORLD, SPORTS, TECHNOLOGY or BUSINESS), a title, a summary and a "meta" line with the author and reading time.
- **Sidebar:** three panels: Featured Stories (three small image-and-text items), Trending Now (an ordered list of five topics) and Our Author (photo, name Priya Sharma, role and a short note).
- **Footer:** the site name, the tagline "Real News · Trusted Views", three links and the copyright line.

4. CSS Source Code (news.css)

The screenshots below show the stylesheet in the code editor, followed by the complete source listing.

Source Code in the Editor

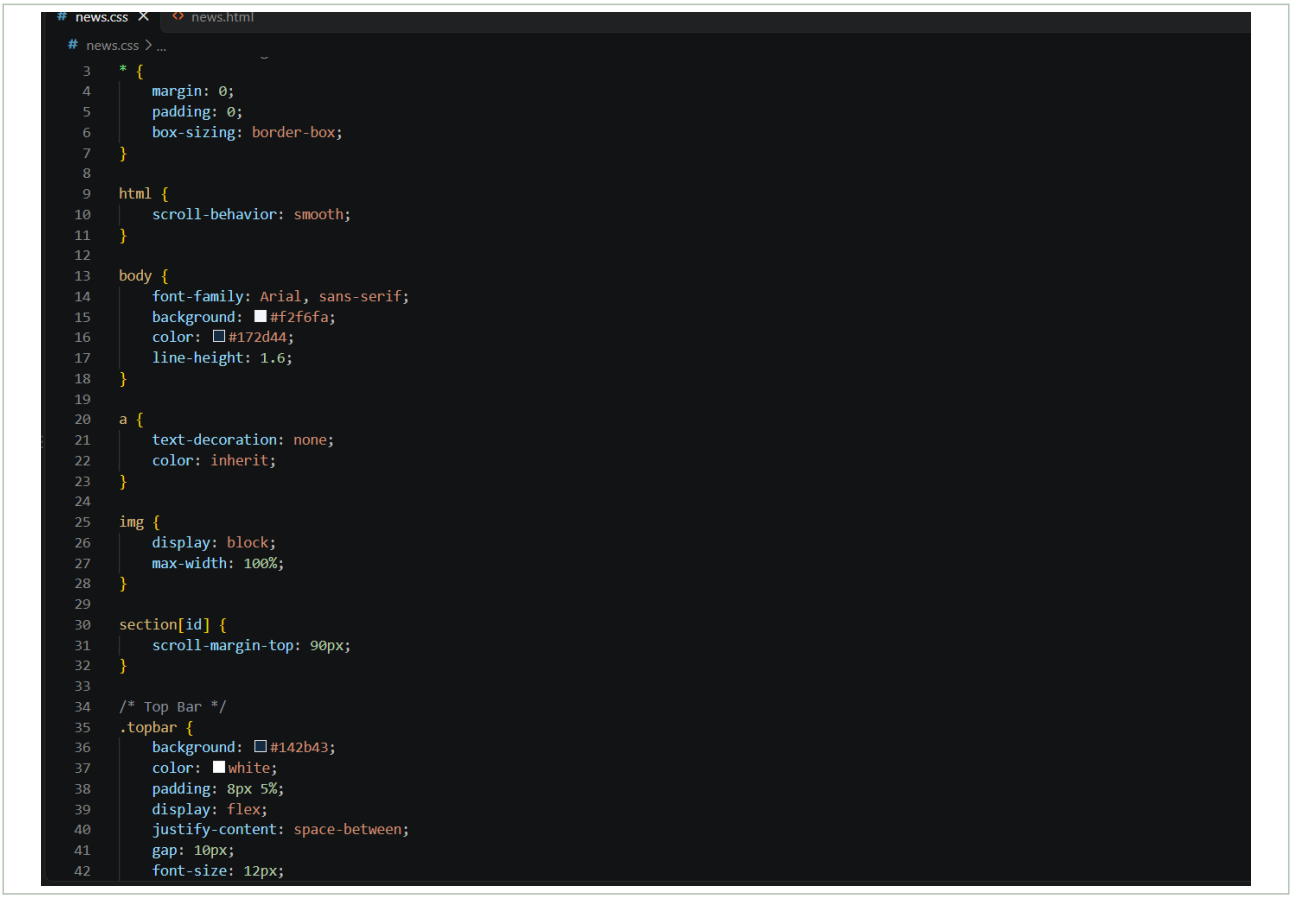


Figure 3: news.css in the code editor - general settings, body, links, images and the top bar (lines 3-42)

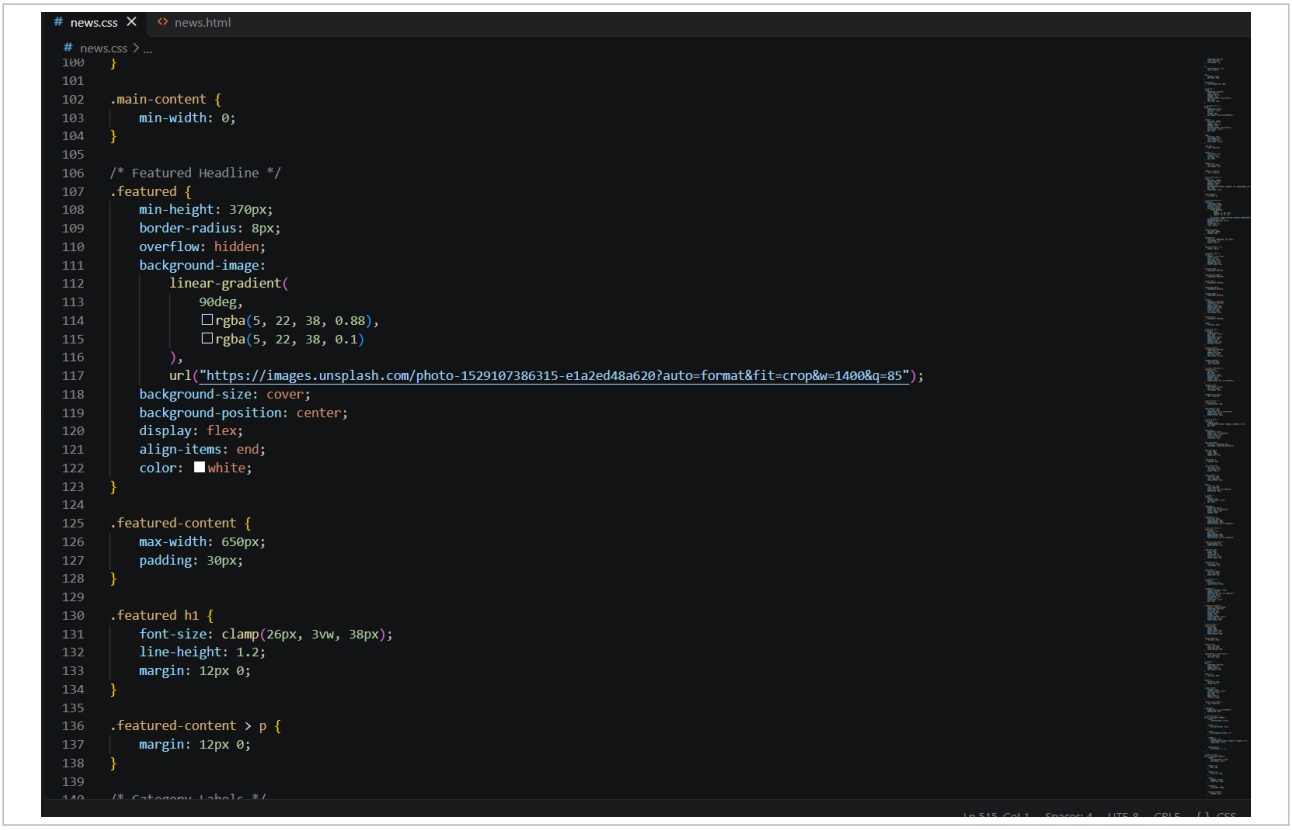


Figure 4: news.css in the code editor - main content and the featured headline with the image overlay (lines 100-139)

Complete Source Listing

```
/* General Settings */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background: #f2f6fa;
    color: #172d44;
    line-height: 1.6;
}

a {
    text-decoration: none;
    color: inherit;
}

img {
    display: block;
    max-width: 100%;
}

section[id] {
    scroll-margin-top: 90px;
}

/* Top Bar */
.topbar {
    background: #142b43;
    color: white;
```

```
    padding: 8px 5%;
    display: flex;
    justify-content: space-between;
    gap: 10px;
    font-size: 12px;
}

/* Navigation Bar */
header {
    background: white;
    position: sticky;
    top: 0;
    z-index: 100;
    box-shadow: 0 2px 8px #00000010;
}

.navbar {
    max-width: 1400px;
    margin: auto;
    padding: 15px 4%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 20px;
}

.logo {
    font-size: 27px;
    font-weight: bold;
    color: #142b43;
    white-space: nowrap;
}

.logo span {
    color: #1674e8;
}

.navbar ul {
    list-style: none;
    display: flex;
    flex-wrap: wrap;
    gap: 20px;
}

.navbar ul a {
    font-size: 14px;
    font-weight: bold;
}

.navbar ul a:hover {
    color: #1674e8;
}

/* Main Page Layout */
.layout {
    max-width: 1400px;
    margin: 20px auto;
    padding: 0 20px;
    display: grid;
    grid-template-columns: minmax(0, 3fr) minmax(250px, 1fr);
    gap: 20px;
    align-items: start;
}

.main-content {
    min-width: 0;
}

/* Featured Headline */
.featured {
    min-height: 370px;
    border-radius: 8px;
    overflow: hidden;
    background-image:
        linear-gradient(
            90deg,
```

```
            rgba(5, 22, 38, 0.88),
            rgba(5, 22, 38, 0.1)
        ),
        url("https://images.unsplash.com/photo-1529107386315-e1a2ed48a620?auto=format&fit=crop&w
            =1400&q=85");
    background-size: cover;
    background-position: center;
    display: flex;
    align-items: end;
    color: white;
}

.featured-content {
    max-width: 650px;
    padding: 30px;
}

.featured h1 {
    font-size: clamp(26px, 3vw, 38px);
    line-height: 1.2;
    margin: 12px 0;
}

.featured-content > p {
    margin: 12px 0;
}

/* Category Labels */
.category {
    display: inline-block;
    color: white;
    font-size: 10px;
    font-weight: bold;
    padding: 4px 9px;
    border-radius: 3px;
}

.national-label {
    background: #0876e8;
}

.international-label {
    background: #8b50d6;
}

.sports-label {
    background: #15965a;
}

.technology-label {
    background: #e63e54;
}

.business-label {
    background: #f07822;
}

.button {
    display: inline-block;
    background: #1674e8;
    color: white;
    padding: 10px 18px;
    border-radius: 5px;
    margin-top: 5px;
    font-size: 14px;
    font-weight: bold;
}

.button:hover {
    background: #0958b8;
}

.author {
    font-size: 12px;
}
/* Breaking News */
```

```
.breaking {
    display: flex;
    align-items: center;
    gap: 15px;
    background: white;
    margin-top: 15px;
    padding: 12px;
    border-radius: 5px;
    overflow: hidden;
}

.breaking strong {
    background: #e53945;
    color: white;
    padding: 6px 10px;
    font-size: 11px;
    white-space: nowrap;
}

.breaking marquee {
    font-size: 13px;
    color: #344c65;
}

/* Category Navigation */
.category-nav {
    display: flex;
    gap: 25px;
    overflow-x: auto;
    background: white;
    padding: 15px;
    margin: 18px 0;
    border-bottom: 2px solid #e1e9f2;
}

.category-nav a {
    white-space: nowrap;
    font-size: 14px;
    font-weight: bold;
}

.category-nav a:hover {
    color: #1674e8;
}

/* News Sections */
.news-section {
    margin-bottom: 35px;
}

.news-section > h2 {
    font-size: 23px;
    border-left: 5px solid #1674e8;
    padding-left: 12px;
    margin-bottom: 16px;
}

/* Article Grid */
.news-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 20px;
}

.news-card {
    background: white;
    border: 1px solid #e0e8f0;
    border-radius: 7px;
    overflow: hidden;
    transition: 0.3s;
}

.news-card:hover {
    transform: translateY(-4px);
    box-shadow: 0 8px 20px #172d4418;
}
```

```
.news-card img {
    width: 100%;
    height: 190px;
    object-fit: cover;
}

.card-content {
    padding: 16px;
}

.card-content h3 {
    font-size: 17px;
    line-height: 1.4;
    margin: 10px 0;
}

.card-content > p {
    font-size: 13px;
    color: #53677d;
    margin-bottom: 12px;
}

.meta {
    font-size: 11px;
    color: #637a92;
    border-top: 1px solid #edf1f5;
    padding-top: 10px;
}

/* Sidebar */
.sidebar {
    display: flex;
    flex-direction: column;
    gap: 20px;
}

.side-panel {
    background: white;
    border: 1px solid #dfe8f0;
    border-radius: 7px;
    padding: 18px;
}

.side-panel h2 {
    font-size: 19px;
    margin-bottom: 15px;
    padding-bottom: 10px;
    border-bottom: 1px solid #e4ebf2;
}

/* Featured Stories */
.side-story {
    display: flex;
    gap: 12px;
    margin-bottom: 18px;
    padding-bottom: 15px;
    border-bottom: 1px solid #e8edf3;
}

.side-story:last-child {
    margin-bottom: 0;
    border-bottom: none;
}

.side-story img {
    width: 95px;
    height: 75px;
    flex-shrink: 0;
    object-fit: cover;
    border-radius: 5px;
}

.side-story h4 {
    font-size: 13px;
    line-height: 1.4;
}
```

```
.side-story p {
    font-size: 10px;
    color: #71839a;
    margin-top: 6px;
}

/* Trending List */
.trending {
    list-style: none;
    counter-reset: trend;
}

.trending li {
    counter-increment: trend;
    padding: 12px 0;
    border-bottom: 1px solid #e7edf3;
    font-size: 13px;
    font-weight: bold;
    display: flex;
    align-items: center;
    gap: 12px;
}

.trending li::before {
    content: counter(trend);
    background: #1674e8;
    color: white;
    min-width: 28px;
    height: 28px;
    display: flex;
    justify-content: center;
    align-items: center;
    border-radius: 50%;
}

/* Author Panel */
.author-photo {
    width: 95px;
    height: 95px;
    border-radius: 50%;
    object-fit: cover;
    margin-bottom: 10px;
}

.author-panel h3 {
    font-size: 17px;
}

.author-role {
    color: #1674e8;
    font-size: 13px;
    margin-bottom: 10px;
}

.author-panel > p:last-child {
    color: #53677d;
    font-size: 13px;
}

/* Footer */
footer {
    background: #142b43;
    color: white;
    padding: 30px 5%;
    text-align: center;
}

footer h2 {
    font-size: 25px;
}

footer p {
    font-size: 13px;
    margin: 5px 0;
}
.footer-links {
```

```
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 25px;
    margin: 20px 0;
    font-size: 13px;
}

.footer-links a:hover {
    color: #8cc3ff;
}

.copyright {
    border-top: 1px solid #43566c;
    padding-top: 15px;
}

/* Tablet Screens */
@media (max-width: 1000px) {
    .navbar {
        flex-direction: column;
    }

    .navbar ul {
        justify-content: center;
    }

    .layout {
        grid-template-columns: 1fr;
    }

    .sidebar {
        display: grid;
        grid-template-columns: repeat(2, minmax(0, 1fr));
        align-items: start;
    }

    .author-panel {
        grid-column: 1 / -1;
    }
}

/* Mobile Screens */
@media (max-width: 600px) {
    .topbar {
        flex-direction: column;
        text-align: center;
    }

    .navbar ul {
        gap: 12px;
    }

    .navbar ul a {
        font-size: 12px;
    }

    .layout {
        padding: 0 12px;
        margin-top: 12px;
    }

    .featured {
        min-height: 380px;
    }

    .featured-content {
        padding: 20px;
    }

    .featured h1 {
        font-size: 27px;
    }

    .news-grid {
        grid-template-columns: 1fr;
```

```
    }

    .news-card img {
        height: 210px;
    }

    .sidebar {
        display: flex;
    }

    .category-nav {
        gap: 18px;
    }

    .breaking {
        gap: 8px;
    }
}
```

5. Description of the CSS Code

How the Stylesheet Works

A CSS rule has the form **selector { property: value; }**. The selector chooses which elements are styled, for example *.news-card* (every element with class "news-card"), *.navbar ul a* (links inside the menu list) or *.news-card:hover* (a card while the mouse is over it). One stylesheet styles the whole page, and a change made in one place is applied everywhere.

General Settings

- *** { margin:0; padding:0; box-sizing:border-box; }** removes the browser's default spacing and makes widths include padding.
- **html { scroll-behavior: smooth; }** makes menu links scroll smoothly to their section.
- **body** sets Arial, a light blue-grey background (#f2f6fa), dark navy text (#172d44) and a line height of 1.6 for easy reading.
- **a { text-decoration:none; color:inherit; }** removes underlines; **img { display:block; max-width:100%; }** keeps images inside their containers.
- **section[id] { scroll-margin-top: 90px; }** is an attribute selector; it leaves space above a section when a link jumps to it, so the sticky menu does not cover the heading.

Key Properties and Techniques Used

Property / technique	Used for	Example from the project
position:sticky; top:0; z-index	Keeps the menu visible while scrolling	header (the menu stays at the top in the output screenshots)
display:grid, grid-template-columns	Page and card layouts	.layout: minmax(0,3fr) minmax(250px,1fr); .news-grid: two equal columns
display:flex, justify-content, align-items, gap	Rows and centred items	Top bar, navbar, breaking bar, sidebar stories
background-image with linear-gradient() and url()	Image overlay	.featured: dark navy gradient from left (0.88) to right (0.1) over a photo
clamp(min, preferred, max)	Text that scales with the window	font-size:clamp(26px, 3vw, 38px) on the headline
border-radius, box-shadow	Rounded corners and soft shadows	Cards, buttons, labels, circular author photo (50%)
object-fit:cover	Crops photos without stretching	Card images (190px high), sidebar images, author photo
border-left, border-bottom, border-top	Accent lines and dividers	Blue bar beside each section heading; thin line above the meta text
colour-coded label classes	Different colour for each category	National blue, International purple, Sports green, Technology red, Business orange
counter-reset, counter-increment, ::before	Automatic numbering inside round badges	.trending li::before { content: counter(trend); }
>, :not, :last-child, [id]	Precise selectors	.news-section > h2, .side-story:last-child
overflow-x:auto, white-space:nowrap	Scrollable row that does not wrap	.category-nav

Image Overlay on the Featured Headline

The featured story uses **two background layers**. The first is *linear-gradient(90deg, rgba(5,22,38,0.88), rgba(5,22,38,0.1))*: a navy colour that is almost opaque on the left and almost transparent on the right. The second is the photograph given by *url(...)*. Because the dark side sits behind the text, the white headline is easy to read while the right side of the photo stays bright. *background-size: cover* and *background-position: center* make the photo fill the box, and *display:flex; align-items:end* places the text at the bottom.

Hover Effects and Transitions

Menu and category links turn blue on hover, the "Read Full Story" button becomes a darker blue, footer links turn light blue, and each news card lifts slightly (*transform: translateY(-4px)*) with a soft shadow. The **transition: 0.3s** on the card makes this change gradual.

Responsive Design with Media Queries

Screen width	Changes made
Up to 1000px (tablets)	Menu stacks below the logo and is centred; the layout becomes one column; the sidebar becomes a two-column grid and the author panel spans the full width (grid-column: 1 / -1).
Up to 600px (phones)	Top bar stacks and is centred; menu text is smaller; less side padding; taller featured box; article cards become a single column with taller images; the sidebar returns to a single column.

6. Output

When news.html is opened in a web browser (with news.css in the same folder), the page appears as a single scrolling web page. The screenshots below show the page from the top to the footer.

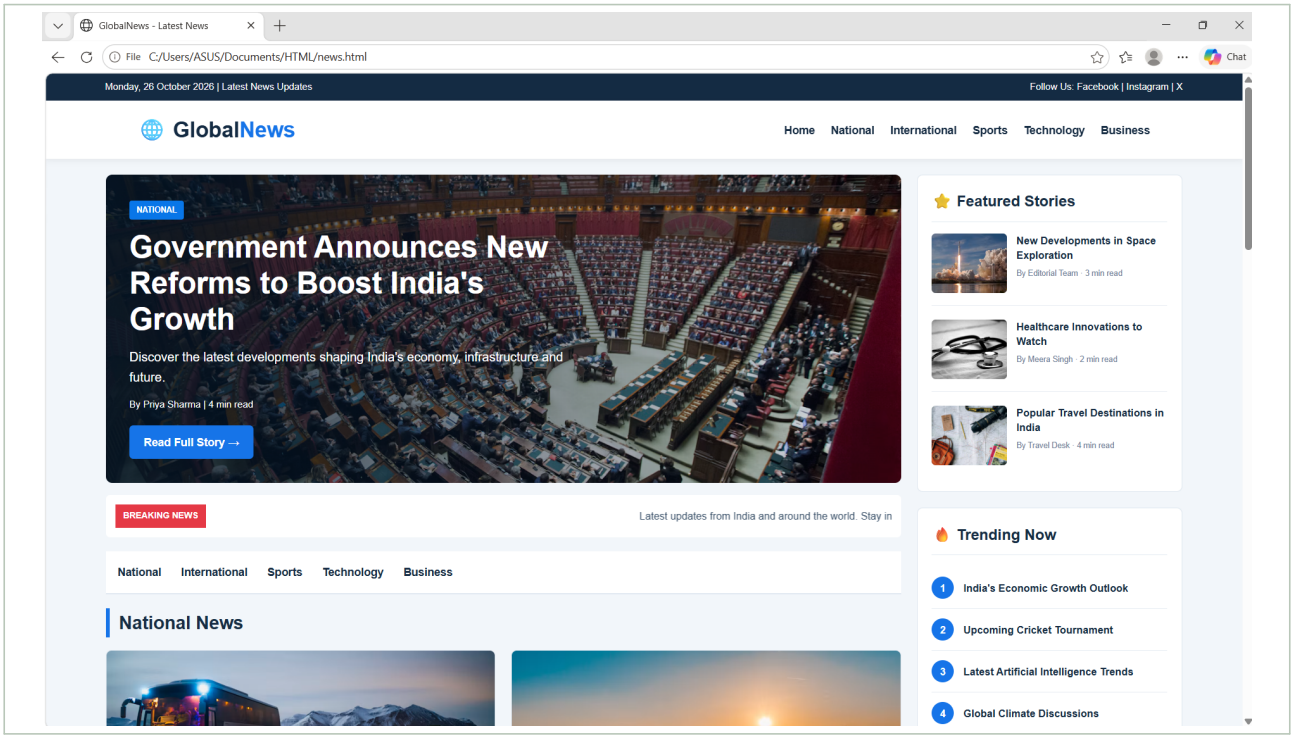


Figure 5: Output - top bar, navigation, featured headline, breaking news bar, sidebar and the start of National News

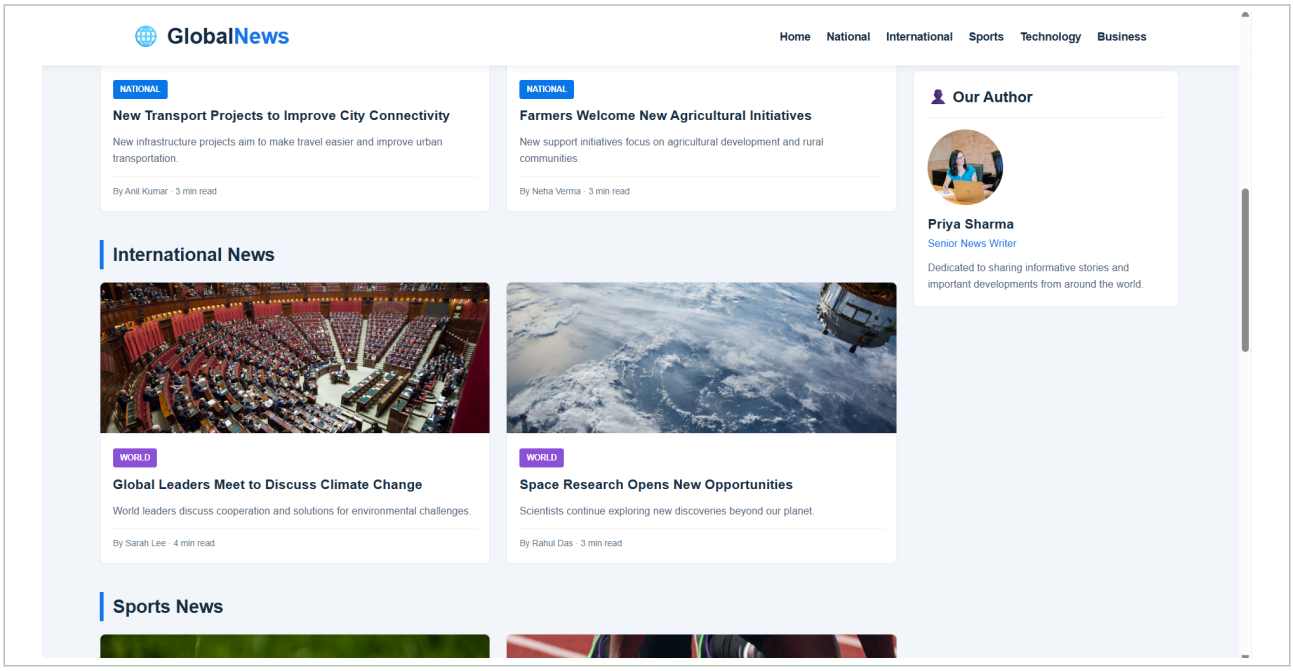


Figure 6: Output - National and International News cards, Our Author panel and the start of Sports News

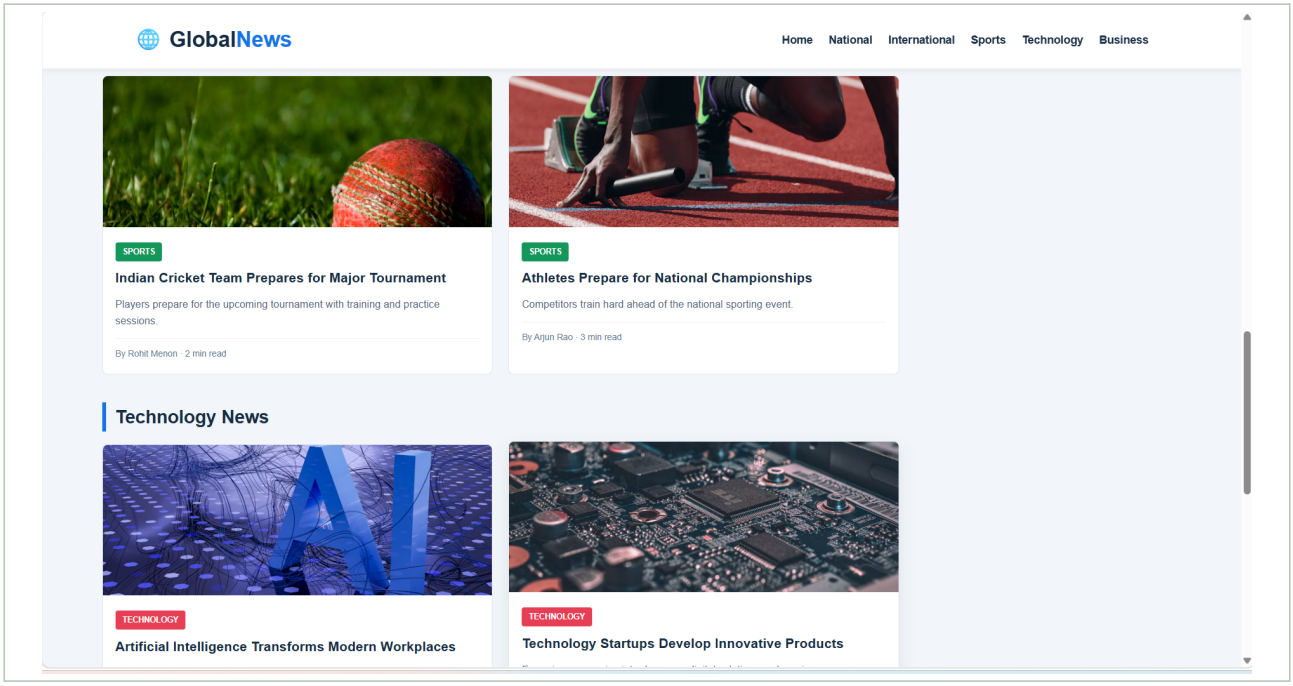


Figure 7: Output - Sports News and Technology News cards

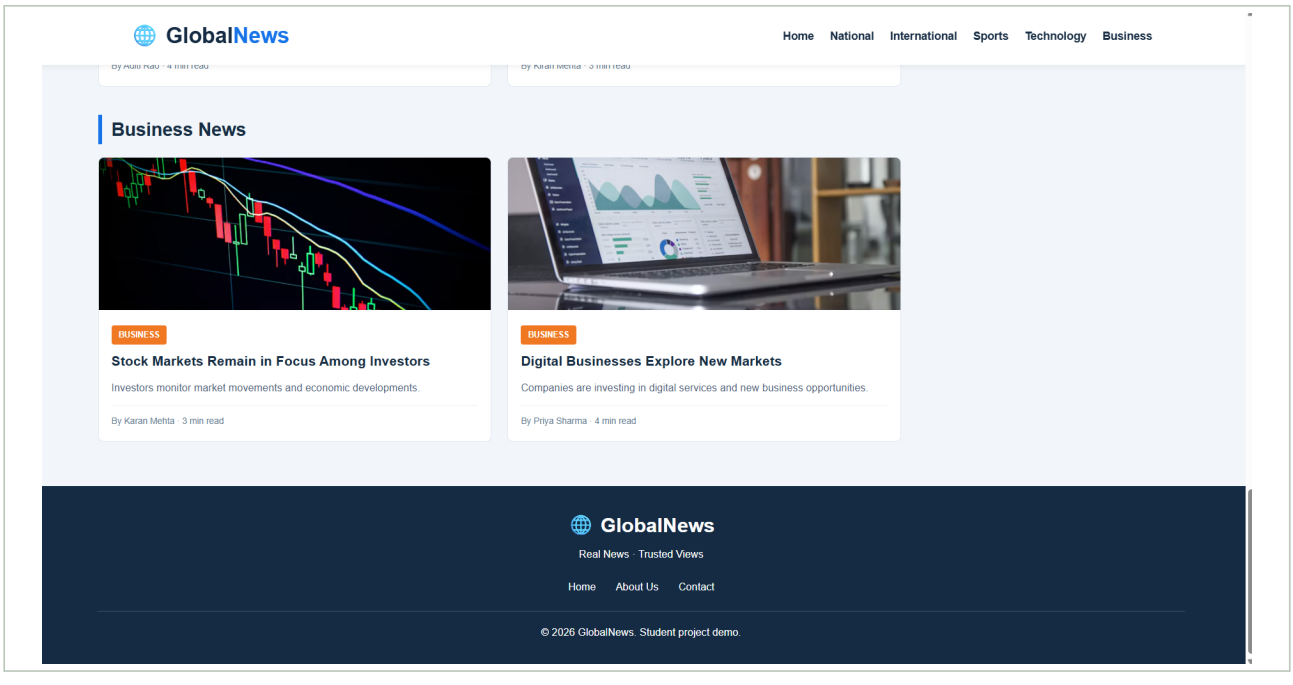


Figure 8: Output - Business News cards and the footer

Description of the Output

- **Top bar and navigation (Figure 5):** a dark navy strip shows the date on the left and the social media names on the right. Below it, a white bar shows the GlobalNews logo with a globe icon and a blue "News", and the six menu links on the right.
- **Featured headline (Figure 5):** a large photo of a parliament hall is darkened on the left by the overlay. A blue NATIONAL label, the white headline, a short paragraph, the author line and a blue "Read Full Story" button sit on top of it.
- **Sidebar (Figures 5 and 6):** white rounded panels to the right of the news. "Featured Stories" lists three stories with small thumbnails; "Trending Now" shows a numbered list

with blue circular badges; "Our Author" shows a round photo with the name Priya Sharma and the role Senior News Writer.

- **Breaking news and category links (Figure 5):** a red BREAKING NEWS label with scrolling text, and a row of five bold category links in a white bar.
- **News sections (Figures 5 to 8):** each section has a heading with a blue bar on its left and two white article cards side by side. Every card has a photo, a coloured label (blue NATIONAL, purple WORLD, green SPORTS, red TECHNOLOGY, orange BUSINESS), a bold title, a short summary and a small line with the author and reading time.
- **Sticky menu (Figures 6 to 8):** the white navigation bar stays fixed at the top of the window while the page is scrolled.
- **Footer (Figure 8):** a dark navy area with the GlobalNews name, the tagline, three links (Home, About Us, Contact) and a centred copyright line below a thin divider.

7. Conclusion

This activity resulted in a complete and responsive news portal home page built with HTML and an external CSS stylesheet. It showed how a large page can be organised into clear parts (menu, headline, news sections, sidebar and footer), and how reusable classes allow the same card design and label styles to be applied many times. Using separate HTML and CSS files also demonstrated how real websites are organised.

Learning Outcomes

- Understood the structure of an HTML document, including DOCTYPE, the lang attribute, meta tags and the title.
- Used semantic tags (header, nav, main, section, article, aside, footer) and lists, headings, images and links correctly.
- Linked an external CSS file to an HTML page and understood why separating content from style is useful.
- Used class and id attributes and wrote CSS rules with class, descendant, child, attribute and pseudo-class selectors.
- Built a two-column page layout and card grids with CSS Grid, and rows and panels with Flexbox.
- Created an image overlay using a layered background with linear-gradient and rgba transparency.
- Made a sticky navigation bar and used scroll-margin-top so that in-page links land correctly.
- Used CSS counters to number a list automatically, and colour-coded labels to group information visually.
- Added hover effects and transitions, and made the page responsive with clamp(), flexible grids and media queries.
- Wrote alt text for images to improve accessibility.

Overall, the activity strengthened problem-solving, attention to design detail and the ability to turn a written requirement into a working, responsive web page, which are skills that form the foundation for further study in web development and software engineering.